Nuclear Fission and Fusion

Answer key with worked solutions

A. Understanding

1. Definitions.

> (a) **Mass defect:** the mass of the separate protons and neutrons minus the mass of the nucleus. **Binding energy:** the energy needed to separate the nucleus into its nucleons, E = Δm c^2. (b) Binding energy per nucleon is **highest near iron-56**. Fission of a heavy nucleus and fusion of light nuclei both produce nuclei **higher on the curve** (more tightly bound); the increase in total binding energy is released.

B. Nuclear equations

2. Completed equations.

> (a) $2\,^1_0\text{n}$, **fission** (236 = 90 + 144 + 2) (b) $^{141}_{56}\text{Ba}$, **fission** (c) ^1_0n, **fusion** (d) ^1_0n, **fusion**

C. Mass and energy

3. Helium-4.

> (a) $\Delta m = 2(1.007276\ \text{u}) + 2(1.008665\ \text{u}) - 4.001506\ \text{u} = \mathbf{0.030376\ u}$
>
> $\Delta m = 0.030376\ \text{u} \times 1.66054 \times 10^{-27}\ \text{kg/u} = \mathbf{5.044 \times 10^{-29}\ kg}$
>
> (b) $E = (5.044 \times 10^{-29}\ \text{kg})(2.998 \times 10^{8}\ \text{m/s})^2 = \mathbf{4.534 \times 10^{-12}\ J}$
>
> (c) $\dfrac{4.534 \times 10^{-12}\ \text{J}}{4\ \text{nucleons}} = \mathbf{1.13 \times 10^{-12}\ J\ per\ nucleon}$

4. 1.00 g of matter.

> $E = (1.00 \times 10^{-3}\ \text{kg})(2.998 \times 10^{8}\ \text{m/s})^2 = \mathbf{8.99 \times 10^{13}\ J}$ (kg m^2/s^2 = J)

© 2026 Chemistry Clarity · chemistryclarity.com · Free for personal and classroom use. Not for resale.

D. Fission and reactors

5. Reactor parts.

> (a) **Moderator** (water or graphite): slows the neutrons so that uranium-235 captures them more easily. (b) **Control rods** (boron or cadmium): absorb neutrons; raising or lowering them controls the rate of fission. (c) **Coolant**: carries heat away from the core to make steam for the turbines. (d) **Critical mass**: the smallest mass of fissile fuel that can sustain a chain reaction.

6. Fission and fusion.

> Fission **splits heavy** nuclei, fusion **joins light** ones; fission produces **long-lived radioactive** waste, fusion mainly helium; fission is used in **power stations** today, fusion is **experimental**. Fusion needs very high temperatures because the nuclei are **positively charged and repel** each other: only very fast collisions bring them close enough to fuse.

7. Energy from uranium-235.

> (a) $200\ \text{MeV} \times 1.602 \times 10^{-13}\ \text{J/MeV} = \mathbf{3.204 \times 10^{-11}\ J}$ per fission
> (b) $3.204 \times 10^{-11}\ \text{J} \times 6.022 \times 10^{23}\ \text{mol}^{-1} = \mathbf{1.929 \times 10^{13}\ J/mol}$
> (c) $\dfrac{1.929 \times 10^{13}\ \text{J/mol}}{235.04\ \text{g/mol}} = \mathbf{8.21 \times 10^{10}\ J/g}$
> (d) Methane: $\dfrac{890\ \text{kJ/mol}}{16.04\ \text{g/mol}} = 55.5\ \text{kJ/g} = 5.55 \times 10^{4}\ \text{J/g}$; ratio $= \dfrac{8.21 \times 10^{10}\ \text{J/g}}{5.55 \times 10^{4}\ \text{J/g}} \approx \mathbf{1.48 \times 10^{6}}$ (about one and a half million times)

E. Fusion and the Sun

8. D–D fusion.

> $\Delta m = 2(2.014102\ \text{u}) - (3.016029\ \text{u} + 1.008665\ \text{u}) = 0.003510\ \text{u}$
> $E = 0.003510\ \text{u} \times 931.5\ \text{MeV/u} = \mathbf{3.27\ MeV}$

© 2026 Chemistry Clarity · chemistryclarity.com · Free for personal and classroom use. Not for resale.

9. The Sun.

(a) $\Delta m = \dfrac{E}{c^2} = \dfrac{3.83 \times 10^{26}\ \text{J}}{(2.998 \times 10^{8}\ \text{m/s})^2} = \mathbf{4.26 \times 10^{9}\ kg}$ each second (J = kg m^2/s^2, so J ÷ (m^2/s^2) = kg)

(b) 1.00 % of the mass: $0.0100 \times 1.989 \times 10^{30}\ \text{kg} = 1.989 \times 10^{28}\ \text{kg}$

$t = \dfrac{1.989 \times 10^{28}\ \text{kg}}{4.26 \times 10^{9}\ \text{kg/s}} = 4.67 \times 10^{18}\ \text{s}$

$4.67 \times 10^{18}\ \text{s} \times \dfrac{1\ \text{yr}}{3.156 \times 10^{7}\ \text{s}} = \mathbf{1.48 \times 10^{11}\ yr}$

© 2026 Chemistry Clarity · chemistryclarity.com · Free for personal and classroom use. Not for resale.